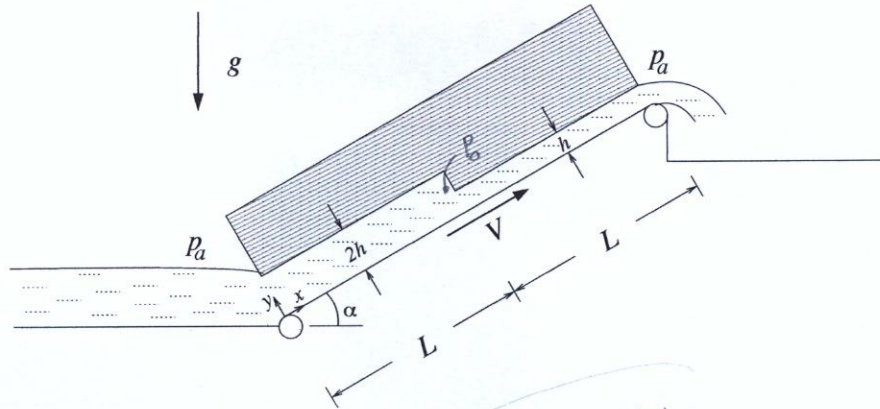


VF9

The conveyor belt of velocity V shown in the figure is used to pump a liquid of density ρ and viscosity μ along a straight inclined channel of angle α including two stretches of equal length L with different heights $2h$ and h , respectively.

- Give the conditions for the liquid-film motion to be dominated by viscosity.
- Compute the pressure distribution along the channel $p(x)$.
- Determine the volume flux Q as well as the minimum value of V for which $Q > 0$.
- Obtain the power needed to drive the conveyor belt.



VISCOSITY DOMINATES IF $\frac{3\sqrt{h}}{\mu} \frac{h}{L} \ll 1$

$$0 < x < L, \quad V_x = +\frac{2h^2}{\mu} P_1 \frac{y}{2h} \left(1 - \frac{y}{2h}\right) + V \frac{2h-y}{2h} \rightarrow Q = \int_0^{2h} V_x dy = \frac{2}{3} \frac{h^3}{\mu} \frac{P_1 - P_0 - \rho g L \sin \alpha}{L} + V h$$

$$L < x < 2L, \quad V_x = \frac{h^2}{2\mu} P_2 \frac{y}{h} \left(1 - \frac{y}{h}\right) + V \frac{h-y}{h} \rightarrow Q = \frac{h^3}{12\mu} \frac{P_2 - P_0 - \rho g L \sin \alpha}{L} + \frac{V h}{2}$$

$$P_2 = P_0 - \rho g L \sin \alpha$$

FROM THE TWO EQUATIONS

$$P_0 - P_a = \frac{2}{3} \frac{\mu V L}{h^2} - \frac{7}{9} \rho g L \sin \alpha$$

$$\text{AND } Q = \frac{5}{9} V h - \frac{4}{27} \frac{\rho g h^3 \sin \alpha}{\mu}$$

$$\rightarrow V_{\min} = \frac{4}{15} \frac{\rho g h^2 \sin \alpha}{\mu}$$

$0 < x < L$:

$$-\frac{\partial(P)}{\partial x} = \frac{P_a - (P_0 + \rho g L \sin \alpha)}{L} \Rightarrow P + \rho g x \sin \alpha = P_a - \frac{P_a - (P_0 + \rho g L \sin \alpha)}{L} x \rightarrow P = P_a + (P_0 - P_a) \frac{x}{L}$$

$$L < x < 2L: -\frac{\partial P}{\partial x} = \frac{P_0 + \rho g L \sin \alpha - (P_a + 2\rho g L \sin \alpha)}{L} \Rightarrow P = P_0 - (P_0 - P_a) \frac{x-L}{L}$$

$$0 < x < L: \tau_1 = \mu \frac{\partial V_x}{\partial y} \Big|_{y=0} = -\frac{\mu V}{2h} + P_1 h$$

$$L < x < 2L: \tau_2 = \mu \frac{\partial V_x}{\partial y} \Big|_{y=0} = -\frac{\mu V}{h} + \frac{P_2 h}{2}$$

$$W = \tau L (\tau_1 + \tau_2) = \tau L \left(-\frac{3}{2} \frac{\mu V}{h} - \frac{3}{2} \rho g \sin \alpha h - \frac{1}{2} \frac{P_0 - P_a}{L} h \right)$$